

Jech 3

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August 2025

1 Cardinal Numbers

Cantor. For any set, $|X| < |P(X)|$. $X \hookrightarrow P(X)$ by $x \mapsto \{x\}$. If $X \xleftrightarrow{f} P(X)$, define $A = \{x \in X : x \notin f(x)\}$. Suppose $A = f(y)$. If $y \in A$, then $y \notin A$, cont. If $y \notin A$, then $y \in A$, cont.

Cantor-Bernstein. If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$. Let $f : A \hookrightarrow B$ and $g : B \hookrightarrow A$. Define recursively together $A_0 = A$, $B_0 = B$, $A_{n+1} = g(B_n)$ and $B_{n+1} = f(A_n)$. Then $\{A_n\}, \{B_n\}$ are descending chains, with $|A_i| = |A|$. We have $A_n - A_{n+1} \longleftrightarrow B_{n+1} - B_{n+2}$ for all $n \in 2N$, $A_{n+1} - A_{n+2} \longleftrightarrow B_n - B_{n+1}$ for $n \in 2N + 1$. There is a bijection $\{A_n - A_{n+1}\} \xleftrightarrow{h} \{B_n - B_{n+1}\}$ st $A_n - A_{n+1} \xleftrightarrow{j} B_m - B_{m+1}$ and $\bigcap_{n < \omega} A_n \xleftrightarrow{f} \bigcap_{n < \omega} B_n$.

If $A \xleftrightarrow{f} A'$, $B \xleftrightarrow{g} B'$ st $A' \cap B' = \emptyset$, then $A \cup B \xleftrightarrow{f \cup g} A' \cup B'$. If $A \xleftrightarrow{f} A'$, $B \xleftrightarrow{g} B'$, then $A \times B \xleftrightarrow{f \times g} A' \times B'$. If $A \xleftrightarrow{f} A'$ and $B \xleftrightarrow{g} B'$, then $A^B \xleftrightarrow{f^g} A'^{B'}$. If $h : B \rightarrow A$, $h \mapsto f \circ h \circ g^{-1}$.

$+$ is associative. If A, B, C are pairwise disjoint, $(A \cup B) \cup C = A \cup (B \cup C)$. $\times$ is associative: $(A \times B) \times C \longleftrightarrow A \times (B \times C)$. $+, \times$ are commutative. $\times$ distributes over $+$: if B, C disjoint, $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

$(\kappa \cdot \lambda)^\mu = \kappa^\mu \cdot \lambda^\mu$. $(A \times B)^C \longleftrightarrow A^C \times B^C$. If $f : C \rightarrow A \times B$, take $f \mapsto (\pi_1 \circ f, \pi_2 \circ f)$.

$\kappa^{\lambda+\mu} = \kappa^\lambda \cdot \kappa^\mu$. $A^{B \cup C} \longleftrightarrow A^B \times A^C$. If $f : B \cup C \rightarrow A$, take $f \mapsto (f|_B, f|_C)$.

$(\kappa^\lambda)^\mu = \kappa^{\lambda \cdot \mu}$. $(A^B)^C \longleftrightarrow A^{B \times C}$. $f : C \rightarrow A^B$ is really $g : C \times B \rightarrow A$.

If $\kappa \leq \lambda$, $\kappa^\mu \leq \lambda^\mu$. $A^C \hookrightarrow B^C$. Take $A \subset B$. If $f : C \rightarrow A$, extend to $g : C \rightarrow B$.

If $0 < \lambda < \mu$, then $\kappa^\lambda < \kappa^\mu$. $A^B \hookrightarrow A^C$. Assume $B \subset C$. If $f : B \rightarrow A$, pick anything to extend to $g : C \rightarrow A$.

Let W be well-ordered, so $W \cong \alpha$ for some $\alpha \in \text{Ord}$, or that $|W| = |\alpha|$. Take α to be minimal. Sps FTSOC $|\alpha| = |\beta|$ for some $\beta < \alpha$. Then $|W| = |\beta|$, cont min of α , so $|W| = \alpha$ is a cardinal.

If $\alpha \in \text{iCard}$, sps $\alpha \notin \text{lOrd}$, so $\alpha = (\alpha - 1) + 1 \rightarrow |\alpha| = |\alpha - 1| \rightarrow \alpha \notin \text{Card}$, cont.

Let $X \subset \text{Card}$ be a set. Then $\alpha = \sup X \in \text{Card}$. First, $\alpha \in \text{Ord}$. Sps FTSOC $\exists \beta < \alpha$ s.t. $\beta \longleftrightarrow \alpha$ and $\beta < \alpha$. By def of sup, $\exists \gamma \in X$ s.t. $\beta < \gamma \leq \alpha$. By ordinals, $\beta \subset \gamma$, so $\beta \hookrightarrow \gamma$. $\beta \longleftrightarrow \alpha$ becomes $\beta \twoheadrightarrow \gamma$, so $\beta \longleftrightarrow \gamma$, so $\beta = \gamma$, cont.

Canonical well-ordering of $\alpha \times \alpha$. Define $<$ on $\text{Ord} \times \text{Ord}$ by $(\alpha, \beta) < (\gamma, \delta)$ if (i) $\max(\alpha, \beta) < \max(\gamma, \delta)$ or (ii) $\alpha < \gamma$ or (iii) $\alpha = \gamma$ and $\beta < \delta$. $<$ is a well-ordering of Ord^2 . Define the injective function $\Gamma : \text{Ord}^2 \rightarrow \text{Ord}$ by $\Gamma(\alpha, \beta)$ the order type of $\{(\alpha', \beta') \in \text{Ord}^2 : (\alpha', \beta') < (\alpha, \beta)\}$.

I claim Γ is surjective. $\emptyset \cong \emptyset$. It can be shown that if $\alpha \in \Gamma(\text{Ord}^2)$, then $\alpha + 1 \in \Gamma(\text{Ord}^2)$. Suppose $\alpha \in \text{lOrd}$ and for all $\beta < \alpha$ there exist (γ, δ) fw $\Gamma(\gamma, \delta) = \beta$. Denote by $A = \{(\gamma, \delta) : \Gamma(\gamma, \delta) < \alpha\}$, so the restriction $\Gamma' : A \rightarrow \alpha$ is bijective increasing. If $(\gamma', \delta') < (\gamma, \delta)$, then $\Gamma'(\gamma', \delta') < \Gamma'(\gamma, \delta) < \alpha$, so $(\gamma', \delta') \in A$. A is a subset of the proper class Ord^2 , so A is an initial segment of Ord^2 , so $\alpha \in \Gamma(\text{Ord}^2)$, so Γ is surjective.

If $J \subset K$ are well-ordered, is it true $\text{type}(J) \leq \text{type}(K)$? We have $J \cong K(x)$, $J \cong K$, or $K \cong J(x)$. Consider the third case. We have the inclusion map $i : J \rightarrow K$ and the order isomorphism $f : K \rightarrow J(x)$, so $g = f \circ i : J \rightarrow J(x)$ is increasing. Suppose FTSOC $A = \{x \in X : g(x) < x\} \neq \emptyset$. Let $x \in A$ be minimal, so $g(g(x)) < g(x) < x$, cont min. Hence, $g(x) \geq x$ for all x , cont.

Suppose $\alpha = \Gamma(\beta, \gamma)$ for $\alpha \in \text{iCard}$. Let $\delta = \max(\beta, \gamma)$. If $\beta < \delta$ and $\gamma = \delta$, then $\text{Ord}^2(\beta, \gamma) = \delta^2 \cup (\beta \times \{\delta\})$. If $\beta = \delta$ and $\gamma < \delta$, then $\text{Ord}^2(\beta, \gamma) = \delta^2 \cup (\delta \times \{\delta\}) \cup (\{\delta\} \times \gamma)$. Finally, if $\beta = \gamma = \delta$, then $\text{Ord}^2(\beta, \gamma) = \delta^2 \cup (\delta \times \{\delta\}) \cup (\{\delta\} \times \delta)$. Let $\kappa = |\delta|$, so $\kappa^2 \leq |\text{Ord}^2(\beta, \gamma)| \leq \kappa^2 + 2\kappa = \kappa^2$, so $|\text{Ord}^2(\beta, \gamma)| = \kappa^2 = |\text{Ord}^2(0, \delta)|$. Note $(0, \delta) \leq (\beta, \gamma)$. If $(\beta, \gamma) > (0, \delta)$, there would be a bijection between $\text{Ord}^2(\beta, \gamma)$ and one of its segments, so $\alpha \notin \text{Card}$. Hence, $\alpha = \Gamma(0, \delta)$.

If $\delta \notin \text{Card}$, then there exists $\delta' < \delta$ fw $\delta \longleftrightarrow \delta'$, so $\delta'^2 \longleftrightarrow \delta^2$, so $\Gamma(0, \delta') \longleftrightarrow \Gamma(0, \delta)$, so $\alpha \notin \text{Card}$ ($\Gamma(0, \delta') \subsetneq \Gamma(0, \delta)$). Hence, $\delta \in \text{Card}$, iCard in fact (since α is infinite). Note $|\alpha| = |\gamma|^2 \geq |\gamma|$, so $\alpha \geq \gamma$. Sps FTSOC there exist γ fw $\alpha > \gamma$ and let α be minimal. Then $|\alpha| = |\gamma|^2 = |\gamma|$, so $\alpha = \gamma$, cont.

It follows if $\alpha \leq \beta$, $N_\beta \leq N_\alpha \cdot N_\beta \leq N_\beta^2 = N_\beta$ and $N_\beta \leq N_\alpha + N_\beta \leq 2N_\beta \leq N_\beta^2 = N_\beta$, so $N_\alpha \cdot N_\beta = N_\alpha + N_\beta = \max(N_\alpha, N_\beta)$.

Let $\{x_\alpha\}_{\alpha \in \text{Ord}}$ be nondecreasing continuous. There exists a fixed point of $\{x_\alpha\}$. Define $y : \omega \rightarrow \text{Ord}$ by $y_0 = 0$ and $y_{n+1} = x(y_n)$. I claim that $y_\omega = \sup y(\omega)$ is

a fixed point. For all $n < \omega$, $x(y_\omega) \geq x(y_n) = y_{n+1}$, so $x(y_\omega) \geq y_\omega$.

If $y(\omega)$ has a maximum element y_m , then $x(y_\omega) = x(y_m) = y_{m+1} \leq y_\omega$. Else, $y(\omega)$ has no maximum, so $y_\omega \in \text{lOrd}$. Note that since $0 \leq y_k$, $\{y_n\}$ is non-decreasing. Now $x(y_\omega) = \lim_{\alpha < y_\omega} x(\alpha) \leq \lim_{n < \omega} x(y_n) = \lim_{n < \omega} y_{n+1} = y_\omega$. Hence, $y_\omega \leq x(y_\omega) \leq y_\omega$, so y_ω is indeed a fixed point.

If $\{x_\alpha\}$ is strictly increasing continuous, there exist arbitrarily large fixed points. Note $x_\alpha \geq \alpha$ for all $\alpha \in \text{Ord}$. The argument is the same, only that $\alpha \leq y_n(\alpha)$ (by induction).

Archimedian principle. Let $\{x_\alpha\}$ be strictly increasing continuous. For all $\beta \in \text{Ord}$, there is a unique α s.t. $x_\alpha \leq \beta < x_{\alpha+1}$. Note $x_{\beta+1} \geq \beta + 1 > \beta$. Let α be minimal s.t. $\beta < x_{\alpha+1}$. Sps FTSOC $\beta < x_\alpha$. If $\alpha \notin \text{lOrd}$, then $\beta < x_{(\alpha-1)+1}$, cont min α . If $\alpha \in \text{lOrd}$, then $\beta < \sup_{\gamma < \alpha} x_\gamma$, so $\beta < x_\gamma < x_{\gamma+1}$ for some γ , cont min α .

Sps $\alpha < \alpha'$ st $x_\alpha \leq \beta < x_{\alpha+1}$ and similarly for α' . Then $\alpha' \geq \alpha + 1$ and these intervals are disjoint, so α is unique. Note the continuity assumption is necessary: $\alpha \mapsto \alpha + 1$ is increasing, but not continuous since $\omega + 1 \neq \sup_{n < \omega} (n + 1) = \omega$. $\alpha + 1 \leq \omega < \alpha + 2 \rightarrow \omega = \alpha + 1$, but $\omega \in \text{lOrd}$, cont.

Let X be a set. There exists an $\alpha \in \text{Card}$ st $|X| < |\alpha|$. Suppose FTSOC for all $\beta \in \text{Ord}$, $\beta \hookrightarrow X$. Denote by $\mathcal{W}$ the set of all well orderings of subsets of X . Then $\text{Ord} \hookrightarrow \mathcal{W} \hookrightarrow P(X \times X)$. If α, β are order isomorphic to the same ordering of a subset A of X , then $\alpha \cong \beta$, so $\alpha = \beta$, so the first map is injective. The second is the inclusion map.

Corollary: There is an uncountable ordinal. $P(N)$ is uncountable, so pick $\alpha > |P(N)|$, so α is uncountable. Pick α to be minimal, which gives ω_1 .

2 Cofinality

$\text{cf } \alpha \leq \alpha$ exists. $\text{cf } \text{cf } \alpha = \text{cf } \alpha$.

If $(\beta \in \text{lOrd}) > 0$, $\text{cf } (\alpha + \beta) = \text{cf } \beta$. $\alpha + \beta = \lim_{\beta' < \beta} (\alpha + \beta') = \lim_{\gamma < \text{cf } \beta} (\alpha + \beta_\gamma)$, so $\text{cf } (\alpha + \beta) \leq \text{cf } \beta$. $\alpha + \beta = \lim_{\gamma < \text{cf } (\alpha + \beta)} (\alpha + \beta)_\gamma = \lim_{\gamma_0 \leq \gamma < \text{cf } (\alpha + \beta)} (\alpha + \beta_\gamma) = \alpha + \lim_{\gamma < \text{cf } (\alpha + \beta)} \beta_\gamma$, so $\beta = \lim_{\gamma < \text{cf } (\alpha + \beta)} \beta_\gamma$, so $\text{cf } \beta \leq \text{cf } (\alpha + \beta)$. I did not check, but it's likely $\text{cf } (\alpha\beta) = \text{cf } (\beta)$ (for $\alpha \neq 0$).

$\text{cf } \omega_{\alpha+1} = \omega_{\alpha+1}$. $\aleph_{\alpha+1} = |\lim_{\beta < \text{cf } \omega_{\alpha+1}} (\omega_{\alpha+1})_\beta| \leq |\text{cf } \omega_{\alpha+1}| \cdot \aleph_\alpha$, so $|\text{cf } \omega_{\alpha+1}| = \aleph_{\alpha+1} \rightarrow \text{cf } \omega_{\alpha+1} \geq \omega_{\alpha+1}$. If $\alpha \in \text{lOrd}$ is nonzero, $\text{cf } \omega_\alpha = \text{cf } \alpha \leq \alpha$ since $\omega_\alpha = \lim_{\beta < \alpha} \omega_\beta$. The exception is $\text{cf } \omega = \omega$.

(Partial) computation of $\text{cf } \alpha$. Write α in normal form. Then

$$\text{cf } \alpha = \begin{cases} \omega & \text{if } \beta_n \notin \text{lOrd}, \\ \text{cf } \beta_n & \text{if } \beta_n \in \text{lOrd}. \end{cases}$$

If $\alpha, \beta < \omega_1$, then $\alpha + \beta, \alpha\beta, \alpha^\beta < \omega_1$ (for the last, for α, β not both 0), so $\epsilon_0 < \omega_1$. Then $\omega^{\omega_1} = \sup_{\alpha < \omega_1} \omega^\alpha \leq \omega_1$, so $\omega^{\omega_1} = \omega$. $\text{cf } \omega_1 = \omega_1$ and $\text{cf } \epsilon_0 = \omega$, so cf is not well-behaved for fixed points of ω^α ?

If $\text{cf } \alpha = \alpha$, if $\beta_n \notin \text{lOrd}$, then $\alpha = \text{cf } \alpha = \omega$. Else, $\alpha = \text{cf } \alpha = \text{cf } \beta_n \leq \beta_n$. Then $\omega^\alpha \leq \omega^{\beta_n} \leq \alpha \rightarrow \omega^\alpha = \alpha$.

If $|A| < |B|$ and $|C| < |D|$, then $|A \times C| < |B \times D|$ (casework on whether A, B, C, D are finite or not).

If $\mathcal{A}$ is a collection of sets s.t. $|\mathcal{A}| < \kappa$ and for all $A \in \mathcal{A}$ has $|A| < \kappa$, then $|\bigcup \mathcal{A}| < \kappa$ is not necessarily true: take $\alpha \in \text{lOrd}$. Note $\bigcup_{\beta < \alpha} \omega_\beta = \omega_\alpha$.

If $\emptyset \subsetneq A \subset \text{Ord}$ is a set with no maximum and $\alpha = \sup A$, then $\text{cf } \alpha \leq \text{type} A$.

Let $\text{type} A \overset{f}{\cong} A$, so $\text{type} A \in \text{lOrd}$. Then $\langle f(\beta) : \beta < \text{type} A \rangle$ is cofinal in α , so $\text{cf } \alpha \leq \text{type} A$.

If